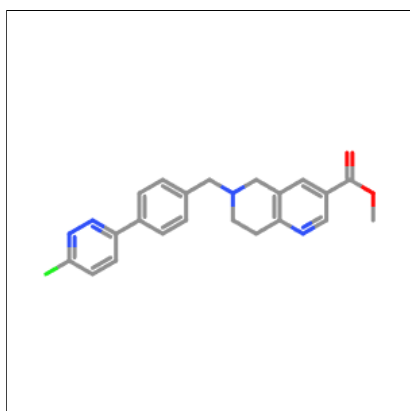
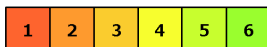


Oral toxicity prediction results for input compound



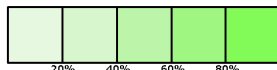
Predicted LD50: 1700mg/kg

Predicted Toxicity Class: 4



Average similarity: 46.95%

Prediction accuracy: 54.26%



Name	COC(=O)c1cnc2c(c1)C[N (Cc1ccc(- c3ccc(Cl)nn3)cc1)CC2
Molweight	395.86
Number of hydrogen bond acceptors	5
Number of hydrogen bond donors	1
Number of atoms	28
Number of bonds	31
Number of rotatable bonds	5
Molecular refractivity	111.25
Topological Polar Surface Area	69.41
octanol/water partition coefficient(logP)	3.69

Toxicity Model Report

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Classification	Target	Shorthand	Prediction	Probability
Organ toxicity	<u>Hepatotoxicity</u>	dill	Inactive	0.67
Organ toxicity	<u>Neurotoxicity</u>	neuro	Active	0.87
Organ toxicity	<u>Nephrotoxicity</u>	nephro	Active	0.52
Organ toxicity	<u>Respiratory toxicity</u>	respi	Active	0.81
Organ toxicity	<u>Cardiotoxicity</u>	cardio	Inactive	0.83
Toxicity end points	<u>Carcinogenicity</u>	carcino	Inactive	0.58
Toxicity end points	<u>Immunotoxicity</u>	immuno	Inactive	0.80
Toxicity end points	<u>Mutagenicity</u>	mutagen	Inactive	0.51
Toxicity end points	<u>Cytotoxicity</u>	cyto	Inactive	0.60
Toxicity end points	<u>BBB-barrier</u>	bbb	Active	0.73
Toxicity end points	<u>Ecotoxicity</u>	eco	Active	0.65
Toxicity end points	<u>Clinical toxicity</u>	clinical	Inactive	0.66
Toxicity end points	<u>Nutritional toxicity</u>	nutri	Inactive	0.71
Tox21-Nuclear receptor signalling pathways	<u>Aryl hydrocarbon Receptor (AhR)</u>	nr_ahr	Inactive	0.77
Tox21-Nuclear receptor signalling pathways	<u>Androgen Receptor (AR)</u>	nr_ar	Inactive	0.97
Tox21-Nuclear receptor signalling pathways	<u>Androgen Receptor Ligand Binding Domain (AR-LBD)</u>	nr_ar_lbd	Inactive	0.96
Tox21-Nuclear receptor signalling pathways	<u>Aromatase</u>	nr_aromatase	Inactive	0.87
Tox21-Nuclear receptor signalling pathways	<u>Estrogen Receptor Alpha (ER)</u>	nr_er	Inactive	0.83
Tox21-Nuclear receptor signalling pathways	<u>Estrogen Receptor Ligand Binding Domain (ER-LBD)</u>	nr_er_lbd	Inactive	0.96
Tox21-Nuclear receptor signalling pathways	<u>Peroxisome Proliferator Activated Receptor Gamma (PPAR-Gamma)</u>	nr_ppar_gamma	Inactive	0.96
Tox21-Stress response pathways	<u>Nuclear factor (erythroid-derived 2)-like 2/antioxidant responsive element (nrf2/ARE)</u>	sr_are	Inactive	0.94
Tox21-Stress response pathways	<u>Heat shock factor response element (HSE)</u>	sr_hse	Inactive	0.94
Tox21-Stress response pathways	<u>Mitochondrial Membrane Potential (MMP)</u>	sr_mmp	Inactive	0.85
Tox21-Stress response pathways	<u>Phosphoprotein (Tumor Suppressor) p53</u>	sr_p53	Inactive	0.83
Tox21-Stress response pathways	<u>ATPase family AAA domain-containing protein 5 (ATAD5)</u>	sr_atad5	Inactive	0.95
Molecular Initiating Events	<u>Thyroid hormone receptor alpha (THRα)</u>	mie_thr_alpha	Inactive	0.89
Molecular Initiating Events	<u>Thyroid hormone receptor beta (THRβ)</u>	mie_thr_beta	Inactive	0.70
Molecular Initiating Events	<u>Transthyretin (TTR)</u>	mie_ttr	Inactive	0.93
Molecular Initiating Events	<u>Ryanodine receptor (RyR)</u>	mie_ryr	Inactive	0.84

Classification	Target	Shorthand	Prediction	Probability
Molecular Initiating Events	<u>GABA receptor (GABAR)</u>	mie_gabar	Inactive	0.57
Molecular Initiating Events	<u>Glutamate N-methyl-D-aspartate receptor (NMDAR)</u>	mie_nmdar	Inactive	0.82
Molecular Initiating Events	<u>alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionate receptor (AMPA)</u>	mie_ampar	Inactive	0.99
Molecular Initiating Events	<u>Kainate receptor (KAR)</u>	mie_kar	Inactive	0.99
Molecular Initiating Events	<u>Achetylcholinesterase (AChE)</u>	mie_ache	Active	0.64
Molecular Initiating Events	<u>Constitutive androstane receptor (CAR)</u>	mie_car	Inactive	0.97
Molecular Initiating Events	<u>Pregnane X receptor (PXR)</u>	mie_pxr	Inactive	0.55
Molecular Initiating Events	<u>NADH-quinone oxidoreductase (NADHox)</u>	mie_nadhox	Inactive	0.95
Molecular Initiating Events	<u>Voltage-gated sodium channel (VGSC)</u>	mie_vgsc	Inactive	0.66
Molecular Initiating Events	<u>Na⁺/I⁻ symporter (NIS)</u>	mie_nis	Inactive	0.79
Metabolism	<u>Cytochrome CYP1A2</u>	CYP1A2	Inactive	0.73
Metabolism	<u>Cytochrome CYP2C19</u>	CYP2C19	Inactive	0.74
Metabolism	<u>Cytochrome CYP2C9</u>	CYP2C9	Active	0.54
Metabolism	<u>Cytochrome CYP2D6</u>	CYP2D6	Active	0.71
Metabolism	<u>Cytochrome CYP3A4</u>	CYP3A4	Inactive	0.60
Metabolism	<u>Cytochrome CYP2E1</u>	CYP2E1	Inactive	0.98

Toxicity targets

Possible binding to toxicity targets is shown below. For more information on the targets, please click on the individual abbreviations.



<u>AA2AR</u>	<u>ADRB2</u>	<u>ANDR</u>	<u>AOFA</u>	<u>CRFR1</u>	<u>DRD3</u>	<u>ESR1</u>	<u>ESR2</u>	<u>GCR</u>	<u>HRH1</u>	<u>NR1I2</u>	<u>OPRK</u>	<u>OPRM</u>	<u>PDE4D</u>	<u>PGH1</u>	<u>PRGR</u>